# AUTOMATIC GENERATION OF BRAIN TUMOR DIAGNOSTIC REPORTS FROM MULTIMODALITY MRI USING LARGE LANGUAGE MODELS

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## ABSTRACT

Early detection and precise characterization of brain tumors are critical for improving patient outcomes and survival rates. Magnetic Resonance Imaging (MRI) remains a cornerstone modality in brain tumor diagnostics, offering high-resolution, multi-sequence imaging essential for capturing the complex structure and pathology of intracranial abnormalities. However, the process of diagnostic report generation is labor-intensive, requiring meticulous review, synthesis, and verification by clinicians. In this study, we present a clinical data pipeline, named iReportMed, designed for automated diagnostic report generation specific to brain tumors. The framework consists of a two-phase model. In the initial phase, multi-modality MRI scans are utilized for comprehensive feature extraction and classification, leading to subject-specific labeling. The second phase employ Large Language Models (LLMs) to produce structured diagnostic reports based on extracted features and identified labels. The proposed framework is initially trained on the UCSF-PDGM dataset and subsequently fine-tuned with real clinical data. Results demonstrate that automated diagnostic reporting is both feasible and potentially advantageous for integration into routine clinical work-flows, offering a promising tool for enhancing efficiency in healthcare settings.

***Index Terms*—** Diagnostic reports, brain tumor, MRI, LLMs

## 1. INTRODUCTION

Automatic generation of diagnosis reports is currently in great demand due to the substantial amount of time required to draft, edit, revise, and approve medical discoveries obtained from medical images into well-reported documents. The common clinical procedure addresses this problem in two phases. The first phase involves medical image processing and feature extraction to reach a diagnosis decision. The second phase involves generating text that describes the findings in detailed, accurate, and concise statements. Artificial intelligence (AI) is now trending as a leading technology with several contributions to different daily applications, achieving remarkable accuracy that is close to human cognition in many clinical applications [1, 2]. Another important leading technology is large language models (LLMs), which demonstrate a significant improvement in generating human-like conversations through publicly accessible online platforms such as ChatGPT and Gemini. In early 2023, a few studies evaluated the performance of chatbots in clinical applications, and the results demonstrated impressive performance, attracting attention to their potential use in the healthcare sector [3]. With the rapid progress of generative AI chatbots, there is controversy regarding the impact of this technology on the future of several healthcare clinical applications. A comparative study evaluated the performance of different LLM models in comparison to medical experts, considering different summarization tasks, including medical reports, and demonstrated the significant performance of the LLMs [4]. Surprisingly, the summarization ability of LLMs outperforms medical experts. Another study evaluated the capabilities of vision LLMs in generating diagnosis reports on computed tomography (CT) images [5]. Results demonstrate that GPT-4V outperformed LLaVa-Med and RadFM in terms of Pearson's correlation with clinicians. Another study also investigated report generation from CT images using LLaMA2-7B [6]. Results obtained from a large-scale dataset demonstrate a remarkable performance improvement in comparison with earlier studies due to the use of a disease-aware attention layer. CopilotCAD demonstrates a framework that fuses foundation models such as MedSAM to enable diagnosis report generation [7]. Results include the performance of using Mistral-7B, Gemma-2B, and TinyLlama-1B as LLMs. ChatCAD+ is proposed as an extension of the earlier version (ChatCAD), with a wide-scope medical report generator and interactive communication through a prompt-enabled environment [8]. With the use of general LLMs such as ChatGPT, ChatCAD+ demonstrates a remarkable ability in high-quality medical diagnosis report generation. A survey on medical LLMs that explores different architectures, applications, and potential use in clinical practice is in [9].

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This study aims at the development of an interactive framework, named iReportMed, for medical image diagnosis using deep learning architectures with LLM assistance. The multi-task image diagnosis models empower extended image knowledge extraction, while LLM technology is used for the generation of expert-like medical diagnosis reports. Without losing generality, the scope of this study is the diagnosis of brain tumors from multi-modality magnetic resonance imaging (MRI). To achieve this goal, we collect data from online resources and real clinical raw data to improve the robustness of the proposed framework. Within this pipeline, two models are combined to generate the diagnosis report. First, a multi-modal deep learning neural network is employed to enable efficient knowledge extraction from MRI at different realization scales. We consider a wider scope that includes abnormality segmentation, target organ annotation, disease diagnosis, and characterization. With the careful design of deep learning architectures customized for mutual-attention multimodality feature extraction, a remarkable improvement in automatic diagnosis is expected. Second, an auto-regressive LLM is used to combine images, image features, and text representing the initial diagnosis to generate a comprehensive medical report. The output of the image diagnosis track will be converted into a text prompt generator to enable communication with chatbots for the automatic generation of diagnosis reports. Up to the best of author's knowledge, this is the first study that proposed a method for automatic report generation of brain tumor from multi-modality MRI.

## 2. MATERIALS AND METHODS

### 2.1. Datasets

In this study, we use a publicly available dataset, named UCSF-PDGM, which consists of 542 subjects with different types of brain tumors [10]. The original MRI scans are acquired using a 3T scanner with different modalities such as T1-weighted, T2-weighted, FLAIR, etc. Images are co-registered and resampled to the 3D space of the T2/FLAIR with 1 mm resolution. The skull is stripped using publicly available software. Tumors are manually annotated using automatic segmentation followed by manual correction by two trained radiologists. A subset of 374 subjects is selected where the diagnosis of Glioblastoma is confirmed. The image size is 240×240×155 pixels ($1mm$ resolution) in NIfTI formats. All images are normalized to reduce inter-subject variability using the normalization method detailed in [11]. Segmentation labels represent different tumor regions such as enhancing tumor, necrotic tumor, and edema. For simplicity, all the three regions are combined as abnormality. The segmentation of parenchyma and the whole brain is available as well. The dataset is associated with brief clinical metadata in CSV formats that is used to generate a synthetic brief diagnosis report using unified prompt as shown in Table 1.

**Table 1**. Synthetic short report generated for UCSF-PDGM dataset sample (Subject ID: UCSF-PDGM-0004).

| |
|---|
| **Prompt:** write a brief medical report for brain tumor case using these parameters: <label>=<value> |
| **Response:** This is a case of a 66-year-old male diagnosed with Glioblastoma, IDH-wildtype, WHO CNS Grade 4. The patient has undergone imaging that reveals a high-grade glioma characterized by necrosis and microvascular proliferation. Molecular and genetic testing indicate the tumor is IDH-wildtype and MGMT promoter methylation status is negative, with an MGMT index of 0, suggesting reduced sensitivity to alkylating agents like temozolomide. The overall survival (OS) is documented as 1303 days. The extent of resection (EOR) achieved was subtotal resection (STR), indicating that not all of the tumor could be surgically removed. No biopsy was performed prior to imaging. Given the aggressive nature of the tumor, a comprehensive treatment plan including radiation and chemotherapy is recommended, along with regular MRI scans to monitor for tumor recurrence or progression. The prognosis remains guarded, and the patient should follow up with neuro-oncology for ongoing care, supportive management, and potential enrollment in clinical trials for novel therapies. |

Another dataset was collected from Suez Canal University Specialized Hospital in DICOM format. The dataset consists of 28 subjects. The slice size is 256×256 (approximately 0.9 mm resolution). The number of slices is not uniform and varies across different sequences (the common axial resolution is about 7 mm). The same data preprocessing procedure is applied to the clinical dataset for consistency. The data are resampled to unify the image size and resolution with the UCSF-PDGM dataset. In this study, we selected four MRI sequences for processing (T1w, T1wc, T2w, and FLAIR). A comprehensive diagnostic report associated with each subject, originally drafted by expert radiologists and confirmed by a senior consultant, is extracted from the hospital archive system. Image and report sharing is approved by the Suez Canal University committee of ethics.

### 2.2. Image Perception Model

The Image Perception Model (IPM) is designed to extract additional features to highlight different regions within an MRI scan, such as the segmented brain region, segmented abnormalities, and the corresponding attention map. The proposed IPM extends the well-known U-Net architecture to handle multi-inputs with mutual attentions mechanism. It incorporates a general score attention layer and multiple encoders instead of a single encoder, enabling feature extraction from multi-modality images with mutual attention. Network architecture and other technical details of the IPM can be found in [11].

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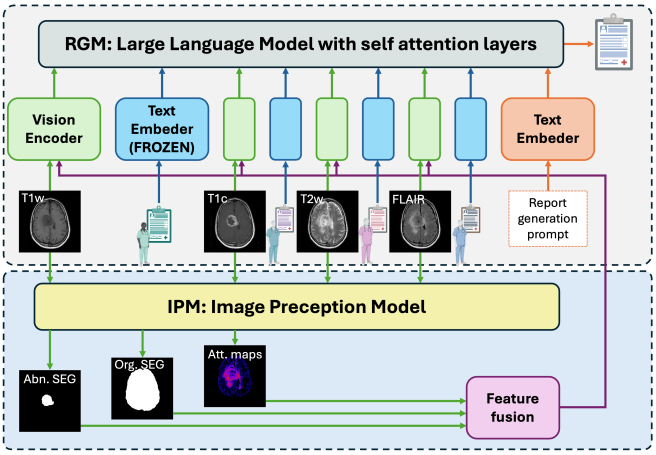


**Fig. 1**. Outline of the proposed data processing pipeline iReportMed. First several MRI images are used to identify tumor and whole brain (segmentation) and attention maps through IPM. Second, output of the first phase is fed along with MRI images and initial diagnosis in auto-regressive RGM in alternative manner. Finally, a report generation prompt is used to generate the final comprehensive report.

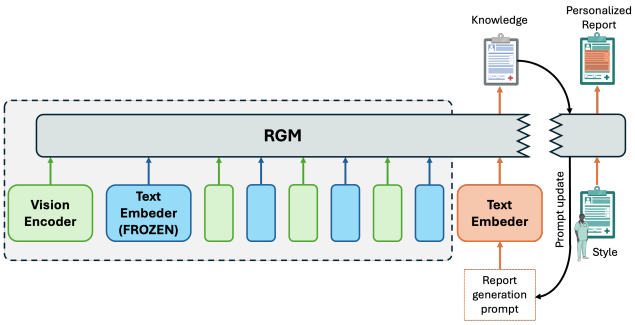


**Fig. 2**. Report personalization process. The initial diagnosis report generated from iRportMed can be customized to fit new style requirements through prompt update process.

### 2.3. Report Generation Model

The Report Generation Model (RGM) is a multimodal few-shot learning architecture. An auto-regressive model with alternative input of images and text is designed to incorporate four MRI sequences with initial diagnosis reports from four different clinical experts. This design enable multimodal settings that enable feature extraction from images to highlight word embedding. Images (MRI scans and IPM output) are encoded using a vision encoder (ViT) [12] as shown in Fig. 1. An open source pre-trained LLM BioMistral-7B is utilized as medical domain trained on PubMed Central [13]. The encoded images and text are embedded into a frozen LLM, allowing the language model to generate comprehensive diagnosis reports using MRI images.

### 2.4. Personalized Reports

To enable personalized diagnostic report the model include an effective scheme for report cutomization as shown in Fig. 2. The comprehensive report generated from iReportMed will be used as knowledge source and will be adjusted using new style to fulfill requirements of report style variability in different healthcare facility and/or report destination. Report customization process is based on updating the report generation prompt considering identification of report knowledge and style. This process is repeated until reaching to adjusted personalized report.

### 2.5. Evaluation metrics and assessment

The accuracy of the IPM is evaluated using segmentation evaluation metrics such as dice coefficient (DSC) and Hausdorff Distance (HD). These metrics are computed for both whole brain and abnormality segmentation. The evaluation of medical diagnosis reports generated by LLMs is a challenging task as requires insightful evaluation from experts. Following the same evaluation method in [4] we requested two medical experts (not included in the initial report data generation) to evaluate output reports in terms of completeness, correctness and conciseness. Each expert independently ranked the reports on a scale from 0 to 10 for each of these three criteria, where 0 indicates the lowest quality and 10 signifies the highest quality. Completeness refers to the extent to which all necessary and relevant information is included in the report, ensuring that it provides a thorough account of the patient's condition. Correctness assesses the accuracy of the medical information presented, verifying that diagnoses, descriptions, and other medical details are precise and reliable. Conciseness evaluates how effectively the information is communicated in a clear and succinct manner, avoiding unnecessary verbosity while maintaining essential details. This ranking system provided a quantitative measure to compare the quality of the reports systematically.

## 3. RESULTS

The IPM is implemented on a workstation with an Intel Xeon (R) CPU at 2.50GHz (28 cores), 128 GB of memory, and an NVIDIA RTX (R) 6000 Ada GPU. A set of 350 subjects from the UCSF-PDGM dataset is used for the evaluation study (245, 70, and 35 subjects are used for training, validation, and testing, respectively). Some subjects are excluded from this study due to improper image quality or observed defects in the segmentation labels. The IPM with four inputs (T1w, T1c, T2w, and FLAIR) slices are used in two training phases for the segmentation of the abnormality and the whole brain. Training is conducted with 100 epochs using a batch size of 2 and the Adam optimizer to optimize the cross-entropy loss function. A single training session is completed within 130 minutes. Evaluation results in comparison with U-Net [14]

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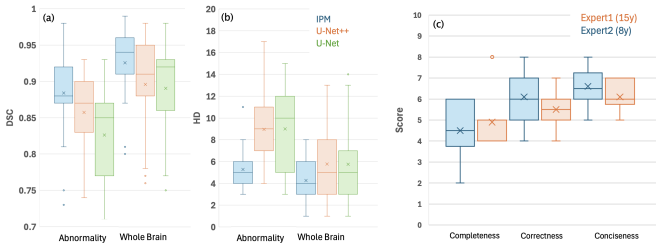


**Fig. 3**. Box plots demonstrating the (a) dice and (b) Hausdorff Distance for abnormality and whole brain segmentation using IPM, U-Net++ and U-Net methods. In (c), evaluation of diagnosis reports generated using iReportMed for 10 subjects by two expert radiologist in terms of completeness, correctness and conciseness.

and U-Net++ [15] are shown in Fig. 3 (a) and (b). It is clear that the performance of IPM demonstrates improvement in segmentation accuracy in terms of DSC and HD values.

The training for RGM is conducted in two phases. In the first phase, 350 subjects from the UCSF-PDGM dataset (four selected modalities) are used along with abnormality segmentation, brain segmentation, and attention maps as image inputs. The corresponding synthesis report inputs are generated similarly to those in Table 1. After the initial training, a fine-tuning session is conducted using 18 randomly selected subjects from the clinical real data. Four different reports are used (each report is used with an inserted statement at the beginning of the report to define the identity of the report editor). Once the training is completed, evaluation is conducted using the remaining 10 subjects. Two external expert radiologists are requested to evaluate the generated reports in terms of completeness, correctness, and conciseness with a grade score of 0-10 by filling out an online form. Evaluation results are shown in Fig. 3 (c). These results demonstrate that there is a slight difference between the two experts, with clear indicators that conciseness is more acceptable compared to correctness and completeness. An ablation study evaluating linguistic understanding is conducted using METEOR and C-BLEU as metrics, demonstrating that the inclusion of an image perception model improves image report quality by 7.2% (METEOR) and 2.8% (C-BLEU).

The model is tasked with generating prompts to match the knowledge in the original report and the style of three different reports: 1) an additional imaging order sent to the radiology department, 2) a simplified family notification report, and 3) a procedure acknowledgment report sent to the accounting department. Examples of personalized reports for single subject are shown in Table 2. Medical experts are requested to highlight text that verifies statements indicating knowledge and those matching the style. Moreover, they are asked to identify non-required statements. Results demonstrate that the further imaging report includes comprehensive knowledge required to be reported to the radiology department. In contrast, reports customized for family notification and the accounting department are quite shallow, with a focus on stylistic statements (Table. 2).

**Table 2**. Personalized version of the report shown in Table 1, styled to match family notification reports. Text color is coded for knowledge and style elements.

| |
|---|
| **Response:** This report is for a 66-year-old man diagnosed with a serious brain tumor called Glioblastoma. The tumor is aggressive and was only partially removed through surgery. It has characteristics that make it less responsive to certain chemotherapy drugs. Given the nature of the tumor, we recommend a comprehensive treatment plan that includes both radiation and chemotherapy, along with regular MRI scans to check for any signs of the tumor coming back or growing. To help us better understand and monitor the tumor, we need to perform several imaging tests: a detailed MRI scan with contrast to see any remaining tumor and changes after surgery, an MRI that looks at the chemical makeup of the tumor, another MRI to check blood flow in the brain, and a special MRI to see if the tumor is affecting the brain's white matter. These tests are crucial for guiding your treatment and ensuring the best possible care. Regular follow-up imaging is essential for managing the patient condition. |

This study has several limitations which we aim to address in future work. One notable limitation is the variability in MRI images caused by the scanner's physical setup and acquisition protocol, which is inconsistent due to the manufacturer's pre-processing techniques that are not always disclosed. This problem can also be mitigated with extra efforts on unification of the image domain with large variety of MRI data. Another limitation is the variability introduced by input from expert radiologists, as their report styles, content, and level of detail can vary significantly in real applications. To mitigate this issue, the proposed framework needs to be trained using a large amount of human input data in additional to clinical data such as biopsy and lab results, which is challenging to acquire in the current stage. Moreover, in clinical usage, it is always recommended to include human-in-the-loop verification to avoid potential model hallucination.

## 4. CONCLUSION

The proposed iReportMed framework represents a promising pipeline towards automating the generation of diagnostic reports from medical images using LLMs. It enables fuse of medical images and text for generating high-quality diagnostic reports. While the results reported here is concise considering the diagnosis of brain tumor, the extension to a plenty of similar applications is direct and pending the availability of training data. In future, we are planning to conduct more comprehensive evaluation studies and extensions to other clinical applications.

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## 5. ACKNOWLEDGMENTS

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**Compliance with ethical standards:** This research study was conducted using human subject data, available in open access and clinical data with prior approval from ethical committee.

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